

PRIMEOR SOLUTIONS

Final Business Insights Report

E-Commerce Data Analytics — Internship Program

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Dataset: E-Commerce | Records: 51,271 | Columns: 21 | Tools: Excel, SQL server, Power BI

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01 EXECUTIVE SUMMARY

This report presents a comprehensive analysis of an E-commerce dataset containing 51,271 order records across multiple markets, product categories, regions, and customer segments. The analysis was performed in three levels - Excel (Data Cleaning & Analysis), SQL (Querying & Insights), and Power BI (Interactive Dashboard) - as part of the Primeor Solutions Data Analytics Internship Program.

TOTAL REVENUE Rs.1,26,35,222 Global sales	TOTAL PROFIT Rs.14,67,498 Net earnings	LOSS-MAKING ORDERS 12,538 24.4% of all orders
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Market-wise Revenue Breakdown

#	Market	Revenue (Rs.)	Share %
1	APAC	35,81,785	28.3%
2	EU	29,37,166	23.2%
3	US	22,97,022	18.2%
4	LATAM	21,63,919	17.1%
5	EMEA	8,05,053	6.4%
6	Africa	7,83,345	6.2%
7	Canada	66,932	0.5%

Segment-wise Sales

Segment	Total Sales (Rs)	% of Total	Key Insight
Consumer	65,03,476	51.5%	Primary customer base — B2C focus needed
Corporate	38,22,954	30.3%	Stable and consistent revenue stream
Home Office	23,08,792	18.3%	Growth potential in remote work era

Task 1 — Data Cleaning Summary

Cleaning Operation	Action Performed	Status
Duplicate Records	Removed all duplicate order rows from dataset	Done
Missing / Null Values	Filled with 'Unknown' or replace the name using order_id	Done
Inconsistent Text Values	Standardized Category, Segment, Ship Mode fields	Done
Date Format Issues	Fixed order_date and ship_date to uniform format	Done
Invalid Numerical Values	Corrected negative/zero values in key columns	Done
Extra Spaces & Formatting	Trimmed whitespace from all text columns	Done
Year Column Added	Extracted year from order_date for trend analysis	Done

Task 2 — Basic Excel Analysis Results

Metric	Value
Total Sales	Rs. 1,26,35,222
Total Profit	Rs. 14,68,087
Average Discount	14.29%
Total Orders	25,029
Total Records	51,271
Profit Margin	11.61%

Top 5 Products by Sales

#	Product Name	Sales (Rs.)
1	Apple Smart Phone, Full Size	86,936
2	Cisco Smart Phone, Full Size	76,441
3	Motorola Smart Phone, Full Size	73,159
4	Nokia Smart Phone, Full Size	71,904
5	Canon imageCLASS 2200 Advanced Copier	61,600

Top 5 Loss-Making Products

#	Product Name	Total Profit (Rs.)	Category
1	Cubify CubeX 3D Printer Double Head Print	-8,879.97	Technology
2	Lexmark MX611dhe Monochrome Laser Printer	-4,589.97	Technology
3	Motorola Smart Phone, Cordless	-4,447.04	Technology
4	Cubify CubeX 3D Printer Triple Head Print	-3,839.99	Technology
5	Bevis Round Table, Adjustable Height	-3,649.89	Furniture

Task 3 — Pivot Table: Region-wise & Category Sales

Region	Sales (Rs.)	Category	Profit (Rs.)	Avg Discount
Central	28,21,740	Technology	6,63,206	13.54%
South	16,00,600	Office Supplies	5,18,476	13.74%
North	12,48,121	Furniture	2,86,404	16.80%
Oceania	10,98,253			
Southeast Asia	8,83,702			

Monthly Sales Trend

Month	Sales (Rs.)	Month	Sales (Rs.)
January	7,72,102.00	July	8,32,816.00
February	5,59,129.00	August	12,11,168.00
March	7,60,168.00	September	14,31,874.00
April	6,99,094.00	October	12,07,507.00
May	8,35,409.00	November	14,67,627.00 (highest)
June	12,53,733.00	December	16,04,595.00 (highest)

Note: November & December are peak sales months — strong Q4 festival season demand.

10 SQL Queries — Results Summary

#	Query	Key Finding
Q1	Top 10 Profitable Products	Canon image CLASS 2200 Copier — Rs.25,200 profit (Highest)
Q2	Top 10 Customers by Sales	Tom Ashbrook tops with Rs.40,489 in total sales
Q3	Region-wise Total Sales	Central region leads with Rs.28,21,740
Q4	Category-wise Avg Profit	Technology has highest average profit per order
Q5	Highest Discount Category	Furniture — 85% discount
Q6	Orders with Negative Profit	12,537 orders (24.4%) show negative profit
Q7	Monthly Sales Trend	December peaks at Rs.15,80,816 —Festival season
Q8	Market-wise Revenue Analysis	APAC market tops revenue at Rs.35,81,785
Q9	Top Performing Sub-Categories	Phones lead with Rs. 17,06,642 in total sales
Q10	Ship Mode Usage Analysis	Standard Class used 30,764 times (59.5%)

Task 3 — Business Insights from SQL Analysis

Q1. Which market generates highest revenue?	APAC — rupees 3,581,785
The APAC (Asia-Pacific) market generates the highest revenue of Rs.35,81,785, followed by EU (Rs.29,37,166) and US (Rs.22,97,022). APAC's dominance is driven by high order volumes from Australia, India, China, and Southeast Asian countries. Strong consumer demand and a large customer base contribute to its top position. Businesses should prioritize APAC for marketing investment and product expansion.	
Q2. Which category is least profitable?	Furniture — only rupees 2,86,125 profit
The Furniture category is the least profitable despite generating significant sales. Two major reasons: (1) It receives the highest average discount of 16.8%, and (2) shipping costs for furniture are substantially higher due to the size and weight of items. Sub-categories like Tables and Bookcases frequently generate negative profit margins. Reducing discounts below 10% could significantly improve profitability.	
Q3. Which shipping mode is most commonly used?	Standard Class — 59.5% (30,764 orders)
Standard Class is by far the most popular shipping mode at 59.5% of all orders (30,764). It is preferred for its cost-effectiveness and wide availability across all regions. Same Day delivery accounts for only 5.2% as it is reserved for urgent/high-priority orders. This distribution reflects good cost-optimization already in place.	

Ship Mode Usage Breakdown

Ship Mode	Order Count	% of Total	Usage Pattern
Standard Class	30,764	59.5%	Most cost-effective — widely preferred by customers
Second Class	10,304	20.1%	Moderate speed and cost balance
First Class	7,504	14.6%	Faster delivery for priority orders
Same Day	2,699	5.2%	Urgent/critical orders only

Top 5 Customers by Sales

#	Customer Name	Total Sales in rupees
1	Tom Ashbrook	40,489
2	Tamara Chand	37,453
3	Greg Tran	35,552
4	Christopher Conant	35,187
5	Sean Miller	35,131

04 LEVEL 3 — POWER BI DASHBOARD OVERVIEW

An interactive 3-page Power BI dashboard was created using the cleaned dataset. The dashboard enables dynamic exploration of sales, profitability, customer behavior, and product performance across markets, categories, and time periods.

Dashboard Pages — Content Overview

Page	Title	Visuals Included
Page 1	Executive Overview	Total Sales KPI Total Profit KPI Total Orders KPI Avg Discount KPI
Page 2	Sales Analysis	Sales by Region (Bar) Sales by Market (Column) Monthly Trend (Table chart) Category Performance (pie)
Page 3	Product & Customer Insights	Top 5 Products (Tree map) Top 5 Customers (Column) Profit by Sub-Category (Bar) Segment Analysis (Pie)

Dashboard Features Implemented

Feature	Details	Status
Slicers / Filters	Months, Region, Sub-category, Order priority, Ship mode, Category, Market, Segment slicers on all pages	Done
KPI Cards	4 KPI cards on Executive Overview page	Done
Interactive Visuals	All charts are cross-filterable by slicer selections	Done
Proper Formatting	Labels, titles, axes, legends properly formatted	Done
Consistent Color Theme	Blue theme used consistently across all 3 pages	Done

Top Performing Sub-Categories (Power BI Insight)

#	Sub-Category	Total Sales (Rs.)	Category
1	Copiers	2,58,174	Technology
2	Phones	2,16,730	Technology
3	Bookcases	1,61,703	Furniture
4	Chairs	1,41,956	Furniture
5	Appliances	1,41,681	Office Supplies
6	Accessories	1,29,591	Technology
7	Storage	1,08,522	Office Supplies
8	Binders	72,433	Office Supplies
9	Papers	59,195	Office Supplies
10	Machines	58,711	Technology

05 KEY BUSINESS INSIGHTS & RECOMMENDATIONS

Revenue & Market Performance

- APAC is the top revenue market (Rs.35,81,785) — primary focus area for business growth.
- EU at Rs.29,37,166 is a strong second — European consumer demand is consistent year-round.
- Overall profit margin is only 11.61% — controlling discounts in Furniture can improve this to ~14-15%.
- Canada market is significantly underperforming at only Rs.66,932 (0.5%) — needs strategic review.

Product & Category Analysis

- Technology category is the most profitable at Rs.4,74,904 — Phones and Copiers should be prioritized.
- Furniture is getting loss Rs. -64,114 due to discount and high shipping costs are the main reasons.
- Canon image CLASS 2200 Advance Copier is highest sale single product at Rs.86,936— promote heavily.

Customer & Segment Insights

- Consumer segment drives 51.5% of total sales — B2C marketing strategies should dominate budgets.
- Top 5 customers contribute significantly — a loyalty program can improve retention.
- Home Office segment (18.3%) has untapped growth potential in the remote-work era.

Shipping & Operations

- 59.5% of orders use Standard Class — good cost-efficient shipping already in place.
- 24.4% of orders (12,538) have negative profit — mainly due to excessive discounts in Furniture.
- Same Day shipping (5.2%) should be restricted to high-value orders only.

Seasonal & Trend Insights

- Q4 (Oct-Dec) is peak season — revenue spikes in November (Rs.14,67,627) and December (Rs.16,04,595).
- Q1 (Jan and Feb) shows lowest sales — targeted marketing campaigns needed in off-peak months.
- Consistent seasonal patterns year-over-year — useful for inventory and sales forecasting.

Top 3 Actionable Recommendations

1	Reduce Furniture category discounts to below 10% Expected Impact: Increase category profit margin by ~30%
2	Focus marketing and inventory on APAC & EU markets Expected Impact: Maximize ROI in highest-revenue zones
3	Discontinue or reprice Cubify CubeX 3D Printer Double Head Print Expected Impact: Eliminate top 2 loss-making products, saving ~Rs.12,720/year

06 CONCLUSION & PROJECT COMPLETION

This internship project provided comprehensive hands-on experience across the complete data analytics pipeline - from raw data cleaning in Excel, to structured business querying in SQL, and finally to interactive BI dashboards in Power BI. The analysis of 51,271 E-commerce records across 7 global markets revealed clear patterns and actionable business insights that can directly improve revenue, reduce losses, and optimize operational efficiency.

Final Project Completion Checklist

Task	Level	Submitted File	Status
Data Cleaning in Excel	Level 1	Cleaned_Dataset.xlsx	Done
Basic Excel Analysis (Sales, Profit, Discount)	Level 1	Excel_Analysis_File.xlsx	Done
Pivot Tables & Charts	Level 1	Excel_Analysis_File.xlsx	Done
SQL Data Import	Level 2	Cleaned_Dataset.csv	Done
10 SQL Queries Written & Executed	Level 2	SQL_Queries_File.sql	Done
SQL Business Insights (3 questions)	Level 2	SQL_Queries_File.sql	Done
Power BI 3-Page Dashboard	Level 3	Power_BI_Dashboard.pbix	Done
Dashboard Features (Slicers, KPIs, Visuals)	Level 3	Power_BI_Dashboard.pbix	Done
Final Business Insights Report	Final	Business_Insights_Report.pdf	Done

Submitted Files

File Name	Format	Level	Status
Cleaned_Dataset	.xlsx	Level 1	Done
Excel_Analysis_File	.xlsx	Level 1	Done
SQL_Queries_File	.sql	Level 2	Done
Power_BI_Dashboard	.pbix	Level 3	Done
Final_Business_Insights_Report	.pdf	Final	Done